

YARENSK, Russian Federation

WMO: 227980

Lat: 62.1717N

Lon: 49.1094E

Elev: 300

StdP: 14.54

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-28.1	-22.5	-33.4	0.8	-28.1	-27.6	1.2	-22.6	15.9	18.6	13.5	17.4	1.6	40	0.632

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.0	82.1	65.6	78.4	63.4	75.1	62.1	68.2	77.9	66.0	74.8	64.1	71.6	5.3	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
65.0	93.4	72.1	63.0	87.1	69.1	60.9	80.8	67.9	32.6	77.6	30.9	74.6	29.4	71.6	82.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4)	14.8	13.1	11.5	DB	-35.2	88.8	5.2	3.6	-38.9	91.4	-41.9	93.5	-44.8	95.5	-48.5	98.1
(5)				WB	-35.4	72.2	5.2	4.3	-39.1	75.4	-42.1	77.9	-45.0	80.4	-48.7	83.5

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	34.5	6.4	12.9	20.9	35.1	46.5	57.7	61.7	57.3	46.9	34.4	21.3	11.7
	DBStd	21.59	14.44	14.12	11.03	8.53	10.14	7.85	6.37	7.21	6.82	7.36	11.97	15.91
	HDD50	6643	1351	1039	904	452	197	16	1	16	135	483	861	1188
	HDD65	11217	1816	1459	1369	898	580	246	140	255	543	947	1311	1653
	CDD50	990	0	0	0	4	88	246	365	244	42	1	0	0
	CDD65	87	0	0	0	0	6	25	39	17	0	0	0	0
	CDH74	967	0	0	0	0	90	329	406	140	2	0	0	0
	CDH80	212	0	0	0	0	18	97	79	18	0	0	0	0
Wind	WSAvg	5.1	4.9	5.4	5.3	5.1	5.7	5.4	5.0	4.7	4.8	5.4	5.2	4.9
Precipitation	PrecAvg	23.6	1.5	1.1	1.2	1.5	1.8	2.4	2.7	2.5	2.2	2.5	2.1	1.9
	PrecMax	29.7	3.0	2.4	2.7	2.7	3.9	4.9	5.0	6.1	4.6	4.5	4.1	4.0
	PrecMin	18.1	0.4	0.0	0.2	0.2	0.6	0.7	0.9	1.2	0.8	0.4	0.4	0.5
	PrecStd	2.6	0.6	0.6	0.6	0.7	0.7	1.0	1.1	1.0	0.9	0.9	0.9	0.8
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	33.3	35.7	44.9	66.4	81.7	88.8	86.4	82.2	72.2	51.9	45.9	35.5
		MCWB	32.9	33.4	36.6	51.0	62.2	68.5	67.7	67.9	60.7	48.2	45.5	34.4
	2%	DB	31.2	33.7	40.0	56.9	75.2	82.3	82.0	77.4	65.9	48.7	37.9	33.2
		MCWB	30.7	32.4	33.8	46.6	57.9	65.3	65.5	65.3	56.4	46.7	37.4	32.2
	5%	DB	27.4	32.3	37.0	51.5	70.3	77.3	78.5	73.7	62.0	46.7	35.4	32.3
		MCWB	26.6	31.2	32.5	42.2	55.0	62.1	63.6	62.9	54.9	45.1	34.6	31.6
	10%	DB	24.1	30.0	34.9	47.3	65.2	72.9	75.4	69.8	58.1	44.6	33.9	30.6
		MCWB	23.3	28.9	32.0	40.0	52.6	60.5	62.6	61.5	53.1	43.1	33.3	29.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	32.8	34.2	37.7	52.4	64.6	71.2	72.0	70.0	61.8	49.7	45.1	34.5
		MCDB	33.0	34.8	43.2	64.5	78.6	82.1	80.6	78.4	67.8	51.2	45.8	35.1
	2%	WB	30.6	32.7	35.0	47.4	60.4	67.7	68.1	67.1	58.5	47.0	37.5	32.6
		MCDB	31.0	33.7	37.9	55.2	70.2	79.7	76.6	74.8	63.6	48.4	37.8	33.0
	5%	WB	26.9	31.3	33.5	43.3	56.7	65.3	65.7	65.0	56.0	45.3	34.9	31.7
		MCDB	27.5	32.4	35.8	48.9	66.9	74.2	74.5	71.3	60.8	46.5	35.2	32.2
	10%	WB	23.3	28.9	32.2	40.8	53.7	62.4	63.8	62.9	53.6	43.3	33.3	29.9
		MCDB	24.1	29.9	34.6	46.1	63.6	70.3	71.8	68.0	57.6	44.6	33.7	30.7
Mean Daily Temperature Range	5% DB	MDBR	11.8	12.0	17.7	16.9	20.3	19.7	19.0	17.2	14.7	8.4	7.9	10.0
		MCDBR	10.2	7.4	18.8	23.4	28.7	27.2	25.2	22.4	24.1	10.6	5.9	6.4
		MCWBR	10.2	6.8	13.9	14.0	14.9	13.1	10.8	11.4	14.8	9.3	5.7	6.6
	5% WB	MCDBR	10.7	7.4	12.8	19.8	24.6	22.8	21.2	19.6	20.7	10.3	5.9	6.0
		MCWBR	10.7	7.1	9.7	12.6	13.9	12.5	10.9	10.8	13.8	9.5	6.0	6.2
Clear-Sky Solar Irradiance	taub	0.249	0.296	0.328	0.359	0.356	0.357	0.398	0.374	0.338	0.314	0.260	0.227	
	taud	1.980	2.052	2.064	2.163	2.431	2.463	2.351	2.398	2.483	2.365	2.075	1.934	
	Ebn at Noon	125	200	240	260	274	275	259	255	245	202	122	73	
	Edh at Noon	12	23	34	38	32	31	34	30	23	17	11	7	
All-Sky Solar Radiation	RadAvg	78	288	674	1122	1531	1658	1670	1141	633	247	73	34	
	RadStd	11	68	119	142	131	160	144	110	85	31	16	5	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (4 neighbors)	+1.08	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page